MOHAN MALLESHAPPA

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EXPERIENCE

Sitetracker ***New Jersey, USA***
Software Development Engineer Intern *May 2024 – May 2025*

- Enhanced **reusable Apex/LWC datatable**, boosting adoption and scalability across **30%** of Sitetracker product features
- Optimized **production tracking** using **advanced SQL queries**, custom **metadata types**, and **server-side pagination**, improving usability and boosting data retrieval speed by **30%** through **indexing** supporting **product lifecycle** and **product design** decisions
- Refactored and tuned **project scheduling engine** using **Apex batch processing**, improving execution performance by **20%**
- Engineered a comprehensive **React Native duty roster module**, introducing a progressive calendar-based job to automate shift assignments, resource allocation, and troubleshooting workflows, projected to improve workforce productivity by **40%**
- Triaged and resolved **35+** customer issues across **QA** and customer orgs, creating detailed **Jira** tickets and **root cause analyses (RCA)**, and delivering code fixes within **SLA** timeframes by following **SDLC** processes, reducing turnaround time by **30%**

NeuroTechR3 ***New Jersey, USA***
Research Assistant | NJIT *Aug 2024 – Jan 2025*

- Built a scalable **Python Flask-based Chatbot API** for **R3THA**, processing **100+** daily document and user queries using **RAG** with **Pinecone** and **NoSQL** via **DynamoDB**, improving **80%** response accuracy while ensuring **HIPAA** compliance data privacy.
- Devised a **LangChain** powered query classification system converting **natural language** to **DynamoDB** aggregation pipelines, reducing response latency by **40%** through intelligent **RAG / NoSQL routing** of user queries.
- Collaborated with **cross-functiona**l teams to develop **responsive graph generation** system using **Matplotlib** for rehabilitation progress visualization, integrating **AWS S3** storage and **metadata** management with **real-time chart** creation capabilities

Cognitive Clouds ***Bengaluru, India***
Full-Stack Software Engineer *Aug 2022 – May 2023*

- Designed cross-platform beauty salon mobile application using **React Native (TypeScript)**, **Node.js,** and **MongoDB**, featuring real-time appointment scheduling, **frontend dashboards**, push notifications and reducing service booking time by **60%**
- Integrated **real-time** messaging functionality using **Quikblox API** and implemented secure payment processing via **Hyperpay Gateway**, ensuring seamless communication and financial transactions while reducing transaction failures by **25%**
- Improved code quality using **Jest.js** for unit testing with **90%+** coverage and reducing production bugs

Freight Deck Technologies ***Bengaluru, India***
Data Engineering Intern *Jan 2022 – Mar 2022*

- Crafted **microservices** based data ingestion logic to capture and handle high-velocity delivery logs using Apache **Spark, Hadoop**
- Restructured **backend APIs** with indexed queries and data pipelines, improving operational visibility into logistics data

SKILLS

Programming Languages: Python, JavaScript, Java, C++, C, Data Structures and Algorithms Optimization, Go, .NET, C#, Swift
Mobile & Web Technologies: React, ReactNative, Node.js, Django, Flask, REST APIs, Kotlin, Apex, LWC, Fast API, NextJS
Database & Cloud: MySQL, MongoDB, Postgres, Pinecone, SOQL(Salesforce), Google Cloud Platform, AWS, Azure
AI & ML Engineering: Machine Learning, Deep learning, GenAI, LLM, NLP, Langchain, PyTorch, Keras, TensorFlow, Alteryx
Tools and DevOps: CI/CD, Docker, Kubernetes, Postman, Linux, Git, Redis, Kafka, Agile, BitBucket, Visual Studio, Unix

EDUCATION

New Jersey Institute of Technology ***New Jersey, USA***
Master of Science in Computer Science *Sept 2023 – May 2025*
Sir M Visvesvaraya Institute of Technology (VTU) ***Bengaluru, India***
Bachelor of Engineering in Computer Science *Aug 2018 – July 2022*

PROJECTS

Healthy Eating Plate Github
Technologies: MongoDB, Express.js, React.js, Node.js

- Developed personalized **meal planning application** generating **500+** nutrition plans and released as an **open source** project
- Used **React-Redux** for state management, enabling smooth interaction, commenting, and routine sharing across components

Payroll Management System Github
Technologies: React, HTML, JavaScript, SQL Server, PHP

- Led the design and development of a comprehensive **payroll management system** serving **100+ administrators** with **algorithmic calculation engine** and **intuitive interface** for real-time updates to bonuses, overtime, and deduction adjustments

Bytematic Inventory Management System
Technologies: React, JavaScript, Spring Boot, Node.js

- Built scalable **SaaS inventory management system** using **JVM-compatible Spring Boot** with core modules for procurement tracking, purchase management, and real-time dashboards, improving business efficiency by **35%** and customer service delivery

LEADERSHIP AND ACHIEVEMENTS

- Won tech quizzes and hackathons, including **Brainwave**, **TechKnow Quiz**, and **Smart India Hackathon** (Regional Round)
- Facilitated students in **Web Systems** coursework as a **Teaching Assistant at NJIT** for 18 months.